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The Effectiveness of Local Fine-Tuned LLMs: Assessment of the Japanese National Examination for Pharmacists

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Abstract

Large Language Models (LLMs) offer great potential for applications in healthcare and pharmaceutical fields. While cloud-based implementations are commonly used, they present challenges related to privacy and cost. This study examined the performance of locally executable LLMs on the Japanese National Examination for Pharmacists (JNEP). Additionally, we explore the feasibility of creating specialized pharmacy models through fine-tuning with Low-Rank Adaptation (LoRA). Text-based questions from the 97th to 109th JNEP were utilized, comprising 2,421 questions for training and 165 for testing. Four distinct locally executable LLMs were evaluated, including the Microsoft phi-4 and the DeepSeek R1 Distill Qwen series. Baseline performance was initially assessed, followed by fine-tuning using LoRA on the training dataset. Model performance was evaluated based on accuracy scores achieved on the test dataset. In the baseline evaluation against the 109th JNEP, accuracy scores ranged from 55.15–76.36%. Notably, the CyberAgent 32B Japanese and DeepSeek R1 Distill Qwen 32B models achieved the passing threshold (approximately 61%). Following LoRA fine-tuning, phi-4 exhibited a performance increase from a baseline of 60.61–66.06%. This study showed that locally executable LLMs are capable of handling specialized pharmacy knowledge tasks comparable to those in the national pharmacist examination. Moreover, we found that fine-tuning techniques like LoRA can significantly enhance model performance, demonstrating the feasibility of creating robust AI models specifically designed for pharmacological applications. These findings contribute to the understanding of implementing secure and high-performing AI solutions tailored for pharmaceutical use.

Introduction

The rapid advancements in artificial intelligence (AI) technology, particularly Large Language Models (LLMs), significantly expand their potential utility as supportive tools across various fields and industries. These developments also influence the social and individual behaviors of citizens. Within the healthcare sector, including medication and pharmaceutical sciences, LLMs have been utilized for several applications, including providing information, offering diagnostic support, and delivering interactive education. These capabilities underscore the transformative potential of AI in enhancing daily healthcare practices. Within pharmacy education, applications such as personalized learning support, material creation, and knowledge assessment are gaining attention, especially in exploring the potential of LLMs (Ali, 2025; Batson & Mara, 2024; Mortlock & Lucas, 2024).

In recent years, numerous studies have evaluated the performance of AI on medical licensing examinations across various countries. According to MedExamLLM, a platform aggregating performance data of LLMs on global medical examinations, as of April 2025, 16 distinct LLMs have been assessed across 198 medical exams in 28 countries. Notably, the Generative Pretrained Transformer (GPT) series demonstrated superior performance (Zong et al., 2024; *Medexamllm.Bioinf.Org.Cn*, n.d.). Furthermore, studies investigating Japan's national medical licensing examination and other medical qualification exams have reported that the Generative Pretrained Transformer (GPT) series achieves a passing standard in many of these tests (Ishida et al., 2024). Within the pharmaceutical domain, a study demonstrated that GPT-4 achieved high accuracy rates on the Japanese National Examination for Pharmacists (JNEP) (Kunitsu, 2023; Sato & Ogasawara, 2024). This indicates substantial potential not only as a powerful tool to support the professional duties of pharmacists but also highlights its viability as a knowledge assessment tool or a benchmark for learning achievement, thereby enhancing pharmacy education.

However, while passing exams is one aspect, the utilization of cloud-based LLMs for AI applications in educational settings presents challenges. This includes the necessity for persistent internet connectivity, safeguarding sensitive information (including student or patient data) against privacy breaches, addressing costs associated with sustained usage, and managing the uncertainty of consistent AI service due to version updates (Ali, 2025; Batson & Mara, 2024; Mortlock & Lucas, 2024).

To address these issues, locally executable LLMs (hereafter referred to as local LLMs), which operate without requiring internet access, are garnering increasing attention. Recently, relatively compact yet high-performance models such as phi-4 and DeepSeek have been developed, demonstrating substantial performance even within standard local computing environments (Abdin et al., 2024; DeepSeek-AI, Guo, et al., 2025; DeepSeek-AI, Liu, et al., 2025). Furthermore, technological advancements in Parameter-Efficient Fine-Tuning (PEFT) techniques—particularly Low-Rank Adaptation (LoRA)—have enabled efficient model tuning, making the construction of domain-specific AI models increasingly feasible (Hu et al., 2021). This may open pathways for developing AI models specialized in fields like pharmacy, potentially enabling concrete educational applications such as adaptive learning systems tailored to individual student comprehension, support tools for instructors creating materials, or knowledge reference tools for use during practical training.

Despite these advancements, there are few reports on AI applications using local LLMs in educational settings. Even focusing on research related to examinations, local LLMs are rarely employed. An analysis of research trends cataloged in MedExamLLM (as of April 2025) reveals a significant disparity: out of 709 studies, 707 utilized cloud-based LLMs, while only two involved the Llama model

published by Meta Platforms, Inc., which were among the few that used local LLMs (*Medexamllm.Bioinf.Org.Cn*, n.d.; Touvron et al., 2023; Zong et al., 2024).

This disparity underscores that while local LLMs may address the limitations of cloud-based models in clinical settings, research on their practical application remains notably insufficient.

This study examines the performance of local LLMs on the JNEP. It further investigates the development of a pharmacy-specialized model via LoRA-based fine-tuning and evaluates the resulting performance modifications. The focus is on assessing the feasibility of creating models capable of handling complex, knowledge-intensive pharmacy tasks within computationally constrained environments (such as school PCs or tablets), ensuring privacy protection.

While the performance of local LLM on JNEP does not exactly replicate the requirements of AI applications in educational practice, there are notable similarities between the domains. Therefore, this work can also be considered a pilot approach to explore novel avenues for AI utilization in clinical practice and contribute to the development of AI systems capable of processing sophisticated specialized knowledge while maintaining rigorous privacy and security standards essential in healthcare.

Material and Method

Data Source: The Japanese National Examination for Pharmacists (JNEP)

This study utilized data from JNEP spanning administrations from 2012 (97th) to 2024 (109th). The JNEP is a national licensing examination administered annually by the Ministry of Health, Labour and Welfare (MHLW) of Japan (*Japanese National Examination for Pharmacists*, n.d.). Following each administration, the MHLW publicly releases examination questions and their corresponding correct answers. The examination is designed to assess the fundamental knowledge and comprehension required for practicing pharmacists. It evaluates a broad spectrum of competencies essential in clinical practice, including the application of pharmaceutical knowledge to real-world scenarios, comprehension of relevant laws and regulations, and adherence to ethical considerations.

Each administration of the JNEP comprises 345 multiple-choice questions presented in a mark sheet format. These questions are distributed across nine distinct subject areas: Physics, Chemistry, Biology, Hygiene, Pharmacology, Pharmaceutics, Pathophysiology/Drug Therapy, Laws/Regulations/Systems, and Practice.

The questions are further organized into three main blocks, each with specific objectives:

1. **Essential Questions (90 questions):** Focus on fundamental knowledge and are typically presented in a single-best-answer format.
2. **Theoretical Pharmacy Questions (105 questions):** Assess theoretical understanding within the pharmaceutical sciences.
3. **Practical Pharmacy Questions (150 questions):** Utilize specific case studies and clinical scenarios to evaluate practical application skills, particularly in pharmacotherapy and patient management.

In addition to knowledge-recall questions, the examination includes calculation problems requiring candidates to derive numerical results from given conditions, as well as questions demanding clinical judgment, such as determining appropriate pharmacotherapy based on presented patient vignettes.

All examination questions are formulated in Japanese. Each question designates either one or two options as correct; a response is considered accurate only if it precisely matches the designated answer(s).

Furthermore, certain questions are presented sequentially, pertaining to a single case description or instructional prompt. Answering subsequent items in such a series may require consideration of the initial context and preceding responses. It is also noteworthy that, following post-examination academic review, certain questions may be deemed invalid or problematic. In such cases, the MHLW may implement corrective measures, such as nullifying the question (officially declaring 'no correct answer') or crediting all candidates who selected specific, predetermined options.

For the 109th JNEP administered in 2024, the established passing criteria required candidates to satisfy all of the following conditions:

1. Achieve a score of 70% or higher on the 'Essential Questions' block.
2. Attain a score of 30% or higher in each of the nine individual subject areas.
3. Obtain a minimum total of 210 correct answers across the entire examination (approximately 60.87% overall accuracy).

4. Select no more than two 'Contraindicated Choices.'

A 'Contraindicated Choice' refers to a response option designated as such if its selection is considered ethically inappropriate, could lead to a violation of laws or regulations, or entails a significant risk of harm to patients or public health. However, the examination administrators do not publicly disclose specific details regarding the content or rationale for these options.

The overall pass rate for the 109th JNEP (2024) was 68.43%, with 9,296 candidates passing out of 13,585 examinees.

Ethical Approval

This study used public examination questions and corresponding answers, which can be obtained from MHLW of Japan. No ethical approval from the institutional review board is required.

Data Preparation

Each JNEP administration comprises 345 questions, and some of them require interpretation of visual elements, such as chemical structure diagrams or anatomical illustrations, in addition to textual information. Those questions necessitating image analysis were systematically excluded from our dataset due to the model's inability to handle image input. Similarly, any questions officially identified by the MHLW as containing flaws in either the question stem or the provided options were also removed prior to analysis.

Furthermore, a specific protocol was applied to handle sequentially linked questions derived from a single case vignette or instructional prompt. If any question within such a series incorporated visual material, the entire sequence of related questions was excluded. This exclusion criterion was implemented because the dependency on image interpretation in one part of the sequence could preclude an accurate assessment of the text-based reasoning capabilities of LLM on subsequent related questions.

The dataset was divided into two parts: questions from the 97th JNEP (2012) to the 108th JNEP (2023) constituted the training set, while questions from the 109th JNEP (2024) formed the test set. In preparing the training data, explanatory texts linked to questions from these administrations were incorporated and sourced from the 'YakugakuQA' resource (*EQUES/YakugakuQA · Datasets at Hugging Face*, n.d.). Questions without a corresponding explanation in this resource were excluded from the final dataset. The final dataset comprised 2,421 questions for training and 165 questions for testing (**Table 1**).

To ensure compatibility with LLM processing, each question was structured into a standardized JSON format. Mathematical equations and chemical notations involving subscripts or superscripts were converted into LaTeX representation within the JSON strings. The adopted JSON structure for each question entry is outlined below (**Scheme 1.**).

LLM Models Evaluated

Four LLM models capable of local execution (*i.e.*, running on local hardware without constant cloud access) were used in this study (**Table 2**).

The phi-4 model achieves high performance with a modest parameter count of 14 billion, owing to its training regimen that incorporates high-quality synthetic data and implements Pivotal Token Search. The model demonstrates strong performance in Science, Technology, Engineering, and Mathematics domains and exhibits multilingual capabilities, including Japanese (Abdin et al., 2024).

The other three models used were the DeepSeek R1 series. The DeepSeek R1 architecture was designed to enhance LLM performance with a focus on strengthening reasoning capabilities. Constructed upon the open-source DeepSeek-V3-Base foundation, its training process integrates Group Relative Policy Optimization, a reinforcement learning technique, with Supervised Fine-Tuning leveraging Chain-of-Thought data. A specific variant evaluated in our study, DSR1-Qwen32B, was derived through knowledge distillation from the foundational DeepSeek-R1 model into the Qwen 32B architecture; this variant is primarily optimized for tasks in English and Chinese (DeepSeek-AI, Guo, et al., 2025; DeepSeek-AI, Liu, et al., 2025). Given that the JNEPs are administered in Japanese, evaluating models specifically enhanced for Japanese language processing was considered crucial. Consequently, two additional variants provided by CyberAgent, Inc., CA-DSR1-32B and CA-DSR1-14B, which were specifically fine-tuned for Japanese, were included in the assessment (*Cyberagent/DeepSeek-R1-Distill-Qwen-14B-Japanese · Hugging Face*, n.d.; *Cyberagent/DeepSeek-R1-Distill-Qwen-32B-Japanese ·*

Hugging Face, n.d.). These Japanese-adapted models are anticipated to exhibit enhanced proficiency in navigating the nuances of Japanese context and linguistic expressions compared to non-Japanese-tuned models.

LoRA: Low-Rank Adaptation of Large Language Models

LoRA represents a parameter-efficient fine-tuning (PEFT) technique designed to facilitate the adaptation of LLMs to specific tasks or domains with high computational efficiency (Hu et al., 2021). Unlike conventional methods that demand extensive computational resources, LoRA offers a cost-effective and resource-efficient approach. It achieves this by augmenting existing models with additional parameters, allowing for specialized adaptations without modifying the architecture of the original model.

For instance, in fields like healthcare, where understanding specific terminologies and workflows is crucial, LoRA allows for the creation of highly accurate yet computationally lightweight models tailored to these domains. Similarly, applications in law and finance have also been reported to be feasible, demonstrating the broad applicability of LoRA across various industries.

The core principle of LoRA involves freezing the original pre-trained model weights, denoted as W_0, and injecting trainable, low-rank matrices to represent the change in weights (ΔW) during adaptation. Formally, let the original weight matrix be as follows (Hu et al., 2021):

$$W_0 \in \mathbb{R}^{d \times k}$$

LoRA approximates the update ΔW using a low-rank decomposition:

$$\triangle W = BA, \qquad where\ B \in \mathbb{R}^{d \times r}\ and\ A \in \mathbb{R}^{r \times k}$$

The crucial aspect for efficiency is that the rank r is significantly smaller than the original dimensions d and k (*i.e.*, $\mathrm{r} \ll \min(d, k)$). Only the matrices A and B are trained during the adaptation process.

Consequently, the forward pass for an adapted layer modifies the original operation $h = Wx$ (where x is the input and h is the output) as follows:

$$h = W_0 x + \triangle Wx = W_0 x + BAx$$

This formulation allows the model to adapt to new tasks by learning only the low-rank update matrices (A and B), drastically reducing the number of trainable parameters compared to fine-tuning the entire W matrix.

Fine-Tuning Methodology

Four LLMs were fine-tuned using the LoRA technique. The training dataset was constructed from questions extracted from the 97th through the 108th JNEPs. To ensure robust model evaluation, the dataset was randomly split into a training set and a validation set in a 9:1 ratio, with 90% of the data reserved for training and 10% used for validating model performance during fine-tuning.

Each model was fine-tuned as a Causal Language Modeling task, where the objective is to predict the next token given a preceding sequence. The implementation leveraged the Trainer class from the Hugging Face Transformers library (*Trainer*, n.d.). To prevent overfitting and optimize training efficiency, an early stopping mechanism was implemented using a custom callback. Key training parameters are summarized in **Table 3**.

Prompt Formulation

Common prompt structures were used to present input data to the LLMs during both inference (for evaluation) and the LoRA fine-tuning. Each prompt began with an instructional header designed to enforce a consistent output format. For the models phi-4 and DSR1-Qwen32B, this header was provided in English: "*The following is a question from the Japanese National Examination for Pharmacists. Please ensure your response is returned strictly in the following format: \"answer\": comma-separated integer(s) (e.g., 1,2,3)*." For models specifically fine-tuned for Japanese (CA-DSR1-14B and CA-DSR1-32B), an equivalent instruction was provided in Japanese.

The core components of the question were appended to this header. These components were extracted from the corresponding JSON entry, as described in the Data Preparation section, and typically included the primary question text (*input.current_problem*), a list of multiple-choice options (*choices*), and relevant background information or case context (*instruction*) when available.

For questions within a multi-step sequence derived from a single scenario, all preceding questions from that sequence (along with their respective correct answers, derived from *previous_problems*) were included before the text of the current question to ensure sufficient context.

During LoRA fine-tuning, the prompt was further augmented with learning targets. Specifically, the ground truth identifier(s) for the correct answer(s) (*output*) and a detailed step-by-step explanation or rationale (*explanation*) were appended to the prompt. This explanation was enclosed within *<think>* tags (*e.g.*, <think>Detailed reasoning steps here...</think>), explicitly guiding the model's learning and reasoning processes during training.

Computational Resources

All computations were conducted on the SQUID supercomputer at the University of Osaka. SQUID is composed of multiple compute nodes, each equipped with two Intel Xeon Platinum 8368 processors based on the Ice Lake architecture, operating at a base frequency of 2.40 GHz and comprising 38 physical cores per processor, for a total of 76 cores per node. Each GPU node is equipped with 512 GB of RAM and eight NVIDIA A100 GPUs. Model inference during evaluation was performed using half-precision floating-point (FP16) to balance computational efficiency and numerical stability. During the LoRA fine-tuning phase, 8-bit quantization was employed to optimize memory usage on the GPUs while maintaining model performance.

Performance Evaluation

The performance of each LLM was assessed by comparing the answers generated to the official answer key for the JNEP provided by the MHLW. A two-step validation process was implemented to verify the accuracy of the LLM-generated responses, drawing on insights from LLM evaluation methodologies in multiple-choice question answering (Molfese et al., 2025). First, automated parsing using regular expressions was employed to systematically extract the answer choice(s) from the output strings of the model. Second, any ambiguities or formatting inconsistencies not resolved by the automated parsing were manually reviewed and verified by the authors.

Each question in the test set was scored dichotomously: 1 point was awarded if the extracted answer(s) exactly matched the official correct answer(s), and 0 points were awarded otherwise. Zero points included cases of incorrect answers, incomplete outputs, or scenarios where a definitive answer could not be reliably parsed from the generation step of a model.

The primary evaluation metric was accuracy, calculated based on performance on the designated test set (comprising 165 questions from the 109th JNEP). Accuracy was determined by dividing the total number of correctly answered questions by the total number of questions in the test set (165), with the result expressed as a percentage.

Results

This study assessed the performance of four locally executable LLMs, phi-4, DSR1-Qwen32B, CA-DSR1-32B, and CA-DSR1-14B, and their LoRA-tuned versions, on the Japanese National Examination for Pharmacists JNEP, focusing specifically on domain-specific knowledge.

Overall Accuracy

The overall accuracy of the baseline models and their LoRA-enhanced counterparts on the 109th JNEP test set (n = 165 questions) are presented in **Table 4**. Baseline accuracies ranged from 55.15% to 76.36%, with the DSR1-Qwen32B achieving the highest score at 76.36%.

For the LoRA fine-tuned models, accuracies ranged from 54.54% to 76.97%, with the LoRA-enhanced DSR1-Qwen32B model demonstrating peak performance at 76.97%. The changes in accuracy after fine-tuning varied across models, ranging from no change (-0.61 percentage points) to a significant improvement of +5.45 percentage points. Notably, three out of the four models showed enhanced performance post-tuning, with phi-4 achieving the most substantial improvement.

Domain-Specific Accuracy Analysis

Table 5 outlines the changes in accuracy across specific subject domains for the 109th JNEP test set after applying LoRA fine-tuning to four AI models.

Several domains demonstrated strong baseline performance around or above the often-considered passing threshold of approximately 60%. These included Biology, Hygiene, Pathophysiology/Drug Therapy, and Practice. Notably, in the Biology domain (n = 9), both CA-DSR1-32B and DSR1-Qwen32B achieved perfect scores, which were maintained post-LoRA. Furthermore, accuracy scores in the Hygiene domain (n = 25) consistently increased for all models after fine-tuning.

In core pharmaceutical science domains demanding specialized knowledge—Pharmacology (n = 27), Pharmaceutics (n = 14), and Pathophysiology/Drug Therapy (n = 26)—LoRA fine-tuning yielded notable improvements for certain models. The phi-4 model showed gains across all three (+11.1%, +14.3%, and +3.8% percentage points, respectively), while CA-DSR1-32B demonstrated a significant increase in Pathophysiology/Drug Therapy (+11.5% points).

Within the Laws/Regulations/Ethics domain (n = 24), models not specifically tuned for Japanese —DSR1-Qwen32B and phi-4—displayed respectable baseline performance (approximately 54% or 13/24 correct) and achieved considerable accuracy gains following LoRA tuning (+16.7% and +12.5% points, respectively).

Conversely, the Chemistry domain (n = 4) revealed highly variable responses to LoRA fine-tuning. While CA-DSR1-14B and DSR1-Qwen32B performed reasonably well at baseline (50% and 100%, respectively), their accuracy dropped substantially (-50.0% points each) after tuning. In contrast, CA-DSR1-32B, which had the lowest baseline accuracy in this domain (25%, 1 correct), experienced a dramatic improvement (+50.0% points, reaching 3 correct answers) post-LoRA.

Discussion

The passing standard for the 109th JNEP was 60.87% (210/345 correct answers). Among the locally executable LLMs assessed in this study, two models—CA-DSR1-32B and DSR1-Qwen32B —met this threshold based solely on their baseline performance. Furthermore, the application of LoRA fine-tuning enabled the phi-4 model to surpass this threshold. Notably, the LoRA-enhanced phi-4 outperformed the baseline accuracy of the larger CA-DSR1-32B model, highlighting the potential of parameter-efficient fine-tuning techniques like LoRA to enhance the performance of smaller models, making it comparable to, or even exceeding that of larger baseline models.

The high accuracy achieved by phi-4, despite its relatively modest size (14 billion parameters), is particularly noteworthy. Small models like phi-4, with their smaller computational footprint, are well-suited for local deployment, offering advantages in resource-constrained environments. Additionally, these local LLMs can operate without continuous internet access, making them highly relevant for offline learning scenarios or settings where student data privacy is paramount. While passing JNEP does not exactly replicate the requirements of AI-driven pedagogical tools or real-world educational practice, the knowledge domains tested share significant similarities with advanced specialized curricula. Our findings demonstrate the feasibility of constructing LLMs capable of handling the complex knowledge required to meet the passing standard of the JNEP using relatively small computational resources, suggesting their potential as effective learning resources or assessment tools within educational programs. These insights provide a foundation for developing advanced local LLMs that meet the privacy and data security requirements common in educational settings while maintaining specialized subject matter expertise, thereby contributing to the development of effective AI-powered educational tools.

Interpretation of Accuracy Results

Interestingly, we found that the DSR1-Qwen32B model, which was not specifically tuned for Japanese, outperformed the Japanese-optimized CA-DSR1-32B model on the JNEP. This observation suggests that fine-tuning a model specifically for the Japanese language does not necessarily guarantee enhanced performance on specialized Japanese examinations. Moreover, the remarkable performance of DSR1-Qwen32B indicates that models primarily trained on non-Japanese data (likely English and Chinese) can still exhibit strong performance on highly specialized Japanese language tasks. This finding aligns with previous research highlighting the inherent multilingual capabilities within LLMs, enabling them to generalize effectively across languages, even within specialized domains(Saito et al., 2024).

A correlation between model parameter count and JNEP performance was observed, with the 32-billion parameter models (*i.e.*, DSR1-Qwen32B and CA-DSR1-32B) generally outperforming the 14-billion parameter models (*i.e.*, phi-4 and CA-DSR1-14B). This finding aligns with established scaling laws in LLMs, suggesting that the larger parameter counts may enhance knowledge acquisition and integration capabilities, which are crucial for multifaceted knowledge tasks like the JNEP(Kaplan et al., 2020). However, this observation should be considered alongside recent advancements in LLM technology, such as model compression, which demonstrate the potential for smaller models to achieve performance parity with larger counterparts(Dantas et al., 2024). Therefore, whether this size-performance trend will persist requires careful ongoing evaluation, particularly when balancing the performance against the significant advantages of smaller models in terms of computational resource efficiency and operational cost.

LoRA fine-tuning generally improved overall accuracy across models. The enhancement was particularly pronounced for the phi-4 model, which, after tuning, surpassed the baseline accuracy of the larger CA-DSR1-32B model. This underscores the potential of LoRA as a viable and effective fine-tuning strategy within the pharmaceutical domain. However, the impact of LoRA was not uniform across all models; DSR1-Qwen32B and CA-DSR1-14B exhibited only marginal changes post-LoRA. This variability aligns with previous findings reporting heterogeneous effects of PEFT methods across different models and tasks(Fomenko et al., 2024). Consequently, optimizing LLM performance in the pharmaceutical domain necessitates the exploration of model-specific adaptation strategies rather than assuming uniform efficacy of any single technique.

Domain-Specific Performance Insights

Analysis by subject domain revealed that all evaluated models demonstrated baseline accuracy exceeding the approximate 60% passing threshold in several areas, including Biology, Hygiene, Pathophysiology/Drug Therapy, and Practice. These domains often involve tasks that predominantly rely on knowledge recall and have significant overlap with general scientific and medical fields, potentially explaining the inherent strengths of LLMs in these areas. Notably, in the Hygiene domain, all models showed improved accuracy scores after LoRA tuning. Given that Hygiene encompasses broad public health, environmental science, and epidemiological topics according to the JNEP syllabus outline, this domain might be particularly amenable to the knowledge transfer effects facilitated by LoRA (*Japanese National Examination for Pharmacists*, n.d.).

In domains demanding core pharmaceutical expertise—Pharmacology, Pharmaceutics, and Pathophysiology/Drug Therapy—the phi-4 model consistently exhibited performance gains post-LoRA (+11.1%, +14.3%, and +3.8% points, respectively). Similarly, the CA-DSR1-32B model showed a notable improvement in Pathophysiology/Drug Therapy (+11.5% points). These results lend further support to previous studies indicating that fine-tuning enhances performance in specialized domains and suggest LoRA is a suitable technique for refining pharmacy-specific knowledge within LLMs (*A Survey of Large Language Models in Medicine: Progress, Application, and Challenge*, n.d.; Alkhalaf et al., 2024; Giuffrè et al., 2024; Nugraha et al., 2024; Yu et al., 2025).

Regarding the Laws/Regulations/Ethics domain, the models not specifically optimized for Japanese (DSR1-Qwen32B and phi-4) exhibited respectable baseline accuracy (approx. 54%) and achieved substantial improvements after LoRA tuning (+16.7% and +12.5% points, respectively). This potentially indicates that these models inherently captured some knowledge relevant to Japan-specific legal and regulatory frameworks, even without explicit tuning. This observation aligns with previous studies demonstrating that augmenting LLMs with country-specific legal information can enhance accuracy in related tasks (Jiang et al., 2025; Tripathi et al., 2024).

Finally, the Chemistry domain presented divergent behaviors following LoRA tuning. While CA-DSR1-14B and DSR1-Qwen32B showed reasonable baseline performance, they experienced a precipitous drop in accuracy (-50.0% points) post-tuning. Conversely, CA-DSR1-32B displayed the opposite effect, with a dramatic enhancement (+50.0% points) from a low baseline. This volatility may be partly attributable to the very small number of Chemistry questions in the test set (n = 4), which limits definitive conclusions on model performance and LoRA effectiveness in this specific domain. This underscores the need for a more granular investigation with a larger question pool.

Limitations and Future Directions

First, the scope of the dataset evaluated was restricted to text-only questions from the JNEP. The models were not assessed on their ability to interpret diagrams, chemical structures, tables, or other visual formats, which constitute a portion of the complete examination. Consequently, the reported accuracies do not fully reflect the comprehensive, multimodal reasoning capabilities required to achieve a passing score comparable to human examinees. Future work should focus on developing and evaluating models capable of handling these diverse question formats.

Second, the application of LoRA fine-tuning inherently involves managing the trade-off between integrating new specialized knowledge and preserving the model's existing knowledge base (Pletenev et al., 2025). This challenge was reflected in our results; while LoRA yielded substantial performance gains in domains where baseline knowledge appeared relatively low (e.g., Laws/Regulations/Ethics), potential performance degradation, possibly due to knowledge interference, was observed in at least one domain with stronger baseline performance but high variability (Chemistry). Addressing this requires further investigation into more sophisticated LoRA implementations, alternative parameter-efficient tuning methods, or adaptive strategies that can better balance knowledge integration and preservation within specific domains.

Finally, the possibility of data contamination or leakage cannot be definitively ruled out. It remains plausible that parts of the JNEP datasets used in our evaluation were inadvertently included in the extensive corpora used for pre-training the base LLMs. Such leakage would potentially inflate the observed performance beyond the true generalization capabilities of the models on genuinely unseen examination content. Ascertaining the precise composition of training data for large foundation models remains a persistent challenge for the field, impacting the interpretation of benchmark evaluations.

Conclusion

This study demonstrates the feasibility of constructing highly educational tools for tasks requiring advanced specialized knowledge, even in environments with limited computational resources, through performance evaluation of local LLMs on JNEP questions. In particular, the results showing that small-scale models achieved accuracy rates comparable to larger models after fine-tuning suggest that secure, high-performance AI solutions can be practically implemented in resource-constrained environments such as university computer labs or on student personal devices.

Furthermore, domain-specific analysis confirmed consistent performance improvements across specialized areas, including pharmacology, pharmaceutics, and pathophysiology/pharmacotherapy after LoRA tuning, revealing the potential for local LLM fine-tuning to enhance expertise in pharmaceutical domains. Meanwhile, the variability observed in performance improvements through LoRA tuning suggests the need for further optimization methodologies, highlighting an important area for future research.

The findings of this study demonstrate that high-performance local LLMs, capable of operating securely within educational environments, can accurately process specialized pharmaceutical knowledge, paving the way for their practical implementation as learning and assessment aids. Future research directions include expanding evaluations to broader pharmaceutical subject areas relevant to the curriculum, exploring specific pedagogical applications (*e.g.*, personalized tutoring systems, automated feedback tools, interactive learning simulations), improving model interpretability and reliability for educational use, and developing a wider range of effective applications in pharmaceutical education.

Declarations

Author contributions

Conceptualization: HA, YST. Data curation: HA, Methodology: HA, YST. Funding acquisition: HA, AH, KI, YST, DT, KF. Software: HA. Investigation: HA. Resources: DT, KF. Writing – Original Draft: HA, YST. Writing – Review & Editing: HA, YST, DT, KF, AH, MO, KI.

Data Availability

The datasets supporting the findings of this study are publicly accessible as follows:

1. Data from the Japanese National License Examination for Pharmacists is available via the Ministry of Health, Labour and Welfare website (https://www.mhlw.go.jp/stf/seisakunitsuite/bunya/kenkou_iryou/iyakuhin/yakuzaishi-kokkashiken/index.html).

2. The YakugakuQA Dataset can be accessed at Hugging Face (https://huggingface.co/datasets/EQUES/YakugakuQA).

The models utilized in this study are hosted on the Hugging Face platform and are publicly available under the following repositories:

1. DeepSeek R1 Distill Qwen 32B Japanese (https://huggingface.co/cyberagent/DeepSeek-R1-Distill-Qwen-32B-Japanese)
2. DeepSeek R1 Distill Qwen 14B Japanese (https://huggingface.co/cyberagent/DeepSeek-R1-Distill-Qwen-14B-Japanese)
3. DeepSeek R1 Distill Qwen 32B (https://huggingface.co/deepseek-ai/DeepSeek-R1-Distill-Qwen-32B)
4. Phi-4 (https://huggingface.co/microsoft/phi-4)

The code and model weights supporting the results of this study are available from the corresponding author upon reasonable request.

Additional information

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Conflict of interest

All authors declare no other competing interests.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Gemini and local DeepSeek to confirm English translation and grammar. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Tables

Table 1. Data Overview from JNPE (97th to 109th)

Domain	Number of Questions	
	Training Set	Test Set
Physics	153	11
Chemistry	43	4
Biology	138	9
Hygiene	345	25
Pharmacology	417	27
Pharmaceutics	314	14
Pathophysiology/Drug Therapy	439	26
Laws/Regulations/Ethics	339	24
Practice	398	25
Total	**2586**	**165**

Table 2. Four LLM models used in this study

Short Forms for Models	Model Full Names	Number of Parameters	Developer
phi-4	Phi-4	14 Billion	Microsoft Corporation
DSR1-Qwen32B	DeepSeek R1 Distill Qwen	32 Billion	High-Flyer
CA-DSR1-32B	DeepSeek R1 Distill Qwen fine-tuned for Japanese	32 Billion	High-Flyer and CyberAgent, Inc
CA-DSR1-14B	DeepSeek R1 Distill Qwen fine-tuned for Japanese	14 Billion	High-Flyer and CyberAgent, Inc

Table 3. Detailed parameters and values used in fine-tuning.

Parameter Category	Parameter Name	Value/Details
LoRA Configuration	Rank (r)	8
	Alpha Scaling Factor (alpha)	16
	Injected Layers	q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj
	Dropout Probability	0.1
Training Parameters	Learning Rate	2×10^{-5}
	Maximum Training Duration	20 epochs
	Learning Rate Warmup	The first 10% of the training steps
	Per-device Batch Zize	2
	Gradient Accumulation Steps	8
	Effective Batch Size (with gradient accumulation)	16
	Maximum Input Sequence Length	4096 tokens
Early Stopping Criteria	Validation Loss Threshold	1.0
	Plateau Detection	Training terminates if no improvement in validation loss is observed across a predefined number of evaluation steps

Table 4. Overall Accuracy Scores on the Test Set (the 109th JNPE)

Model	Accuracy (%)		
	Base Model	LoRA Fine-tuned Model	$\Delta_{Fine-tuned-base}$
CA-DSR1-14B	55.15	54.54	-0.61
CA-DSR1-32B	64.24	67.88	3.64
DSR1-Qwen32B	76.36	76.97	0.61
phi-4	60.61	66.06	5.45

Table 5. Domain-Specific Accuracy and Changes Attributable to LoRA Fine-Tuning on the Test Set (the 109th JNPE)

Domain	n	CA-DSR1-14B			CA-DSR1-32B			DSR1-Qwen32B			phi-4		
		Base (n)	Fine-tuned (n)	Δ(%)	Base (n)	Fine-tuned (n)	Δ(%)	Base (n)	Fine-tuned (n)	Δ(%)	Base (n)	Fine-tuned (n)	Δ(%)
Physics	11	7	6	-9.1	9	8	-9.1	8	9	9.1	8	7	-9.1
Chemistry	4	2	0	-50.0	1	3	50.0	4	2	-50.0	3	3	0.0
Biology	9	7	9	22.2	9	9	0.0	9	9	0.0	6	6	0.0
Hygiene	25	15	16	14.0	18	19	4.0	20	22	8.0	15	17	8.0
Pharmacology	27	10	8	-7.4	17	17	0.0	23	20	-11.1	11	14	11.1
Pharmaceutics	14	7	7	0.0	7	8	7.1	9	8	-7.1	7	9	14.3
Pathophysiology/Rx	26	16	15	3.8	18	21	11.5	24	22	-7.7	19	20	3.8
Laws/Regulations/Ethics	24	12	13	4.2	12	12	0.0	13	17	16.7	14	17	12.5
Practice	25	15	16	4.0	15	15	0.0	16	18	8.0	17	16	-4.0

Scheme

Scheme 1 is available in the Supplementary Files section.

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- scheme1.docx